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Introduction to Week 2

Lesson 1: Linear Algebra Review

Lesson 2: Linear Regression and Statistics

Lesson 3: Confidence Interval & Hypothesis Testing

Video: Confidence Interval 5 min

Video: Hypothesis Testing 8 min

Graded Assignment: Learning Check - Hypothesis Testing Submitted

Weekly Assignments

Your grade: 100%

Your latest: 100%

Your highest: 100%

Next item →

Learning Check - Hypothesis Testing

Review Learning Objectives

Instructions

1. Refer to the following table for Questions 1-5.

1 / 1 point

	Coefficient	Standard Deviation	t-statistic
Intercept	2.937	0.3115	9.42
TV	0.046	0.0014	32.81
radio	0.006	0.0086	21.89
newspaper	-0.001	0.0059	-0.18

The table above represents the output of a linear regression problem in which $\hat{y}$ is the predicted amount of sales (the output) using three predictors (the inputs): TV, radio, and newspaper advertising. From the table above, what conclusions can be drawn at a 95% confidence level?

☐ The Intercept coefficient is significant to predict sales in this regression and should be included in our model because its coefficient is greater than 2. In other words, we can reject the null hypothesis $\beta_0 = 0$.

☒ TV advertising spend is a significant predictor and should be included in our model because the magnitude of its t-statistic is greater than 2.

☒ Correct

☐ We fail to reject the null hypothesis that $\beta_0 = 0$. Upon inspection, we conclude that the radio predictor can be removed from the model.

☒ Under the null hypothesis that $\beta_{\text{radio}} = 0$, we have strong statistical evidence to reject the null hypothesis and conclude that radio advertising does affect sales.

☒ Correct

☒ The 95% confidence interval for the newspaper predictor's coefficient is roughly $[-0.0128, 0.0108]$.

☒ Correct

2. For Questions 2-5, assume that nature behaves according to the following linear model:

1 / 1 point

$$Y = \beta_0 + \beta_1 X + \epsilon$$

where ϵ is a Gaussian random variable following a normal distribution $N(0, \sigma^2)$. Using this model, we obtain the following training dataset:

$$D_T = \{(x_i, y_i)\}_{i=1}^5 = \{(-2, -7/2), (-1, -3), (0, 0), (1, 3), (2, 7/2)\}$$

What is $\hat{\beta}_1$? (Input answer as a whole number.)

2

☒ Correct

$$\hat{\beta}_1 = 0, \hat{\beta}_0 = 0, \text{ so } \hat{\beta}_1 = \frac{-2(-7/2) + (-1)(-3) + 0(0) + 1(3) + 2(7/2)}{1 + 1 + 1 + 1 + 1} = \frac{7 + 3 + 3 + 7}{4} = \frac{20}{4} = 5$$

3. What is $\hat{\beta}_0$? (Answer as a whole number.)

1 / 1 point

0

☒ Correct

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 0$$

4. Compute the RSS of the linear model. (Answer as a real number with one decimal point.)

1 / 1 point

2.5

☒ Correct

Using the above estimates, we have that $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i = 0 + 5x_i$.

Using the values in our dataset, we obtain that

$$RSS = \sum_{i=1}^5 (y_i - \hat{y}_i)^2 = \frac{1}{4} + 1 + 0 + 1 + \frac{1}{4} = \frac{3}{2}$$

5. Compute the RSE of the model. (Answer as a real number with one decimal point.)

1 / 1 point

0.9

☒ Correct

$$RSE = \sqrt{\frac{RSS}{N-p-1}} = \sqrt{\frac{RSS}{N-2}} = \sqrt{\frac{5/2}{5-2}} = \sqrt{\frac{5}{6}}$$